

Financial Time Series Analysis

Lecture 4

- ▶ TA Session Friday April 2, 2021, 11:00am - 12:00pm (EDT)
- ▶ Homework assignment due dates are being moved to Thursday

Outline

- ▶ Weak Stationarity of $AR(p)$
- ▶ Partial autocorrelation function (PACF)
- ▶ Basic time series models (AR, MA, ARMA)
- ▶ Simulating and fitting AR/MA/ARMA using statsmodels

Weak Stationarity of AR(p): 1

Consider an AR(p) model

$$X_t = \phi_0 + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \epsilon_t,$$

where the ϵ_t process is $WN(0, \sigma^2)$.

► If $E(X_t) = \mu$, $\forall t$, then

$$\mu = \phi_0 / (1 - \sum_{i=1}^p \phi_i), \quad \sum_{i=1}^p \phi_i < 1.$$

► Now consider $Y_t = X_t - \mu$.

Weak Stationarity of AR(p): 2

$$\begin{aligned} Y_t &= \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \epsilon_t \\ Y_{t-1} &= Y_{t-1} \\ &\cdot \quad \cdot \quad \cdot \\ &\cdot \quad \cdot \quad \cdot \\ Y_{t-p+1} &= Y_{t-p+1} \end{aligned}$$

If we define $\mathbf{Y}_t = (Y_t, \dots, Y_{t-p+1})^T$, $\epsilon_t = (\epsilon_t, 0, \dots, 0)^T$ and

Weak Stationarity of AR(p): 3

$$\mathbf{M} = \begin{bmatrix} \phi_1 & \phi_2 & \cdots & \phi_{p-1} & \phi_p \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix},$$

then the resulting AR(p) model now becomes in matrix form

$$\begin{aligned} \mathbf{Y}_t &= \mathbf{M}\mathbf{Y}_{t-1} + \epsilon_t \\ &= \mathbf{M}(\mathbf{M}\mathbf{Y}_{t-1} + \epsilon_{t-1}) + \epsilon_t \\ &\dots \\ &= \mathbf{M}^t\mathbf{Y}_0 + \sum_{i=0}^{t-1} \mathbf{M}^i\epsilon_{t-i}, \end{aligned}$$

Weak Stationarity of AR(p): 4

- ▶ For weak stationarity, we need $\mathbf{M}^t \rightarrow 0$ as $t \rightarrow \infty$. This requires that the eigenvalues of $\mathbf{M}$ have modulus strictly less than 1.
- ▶ The eigenvalues can be characterized as the roots of a p-dimensional polynomial

$$\lambda^p - \sum_{i=1}^{p-1} \lambda^{p-i} \phi_i = 0,$$

and the p roots of this equation must have modulus less than 1.

Weak Stationarity of AR(p): 5

- ▶ Alternatively one can define $z = 1/\lambda$ and rewrite the above equation as

$$1 - \sum_{i=1}^p z^i \phi_i = 0.$$

- ▶ This is the autoregressive polynomial, and for weak convergence, its roots have to have modulus greater than 1.

Statsmodels Example: 1

From statsmodels.tsa.arima_process.ArmaProcess

```
import statsmodels.api as sm
```

```
import numpy as np
```

```
arparams = np.array([.75,-.25])
```

```
ar = np.r_[1,-arparams]
```

```
arma_process = sm.tsa.ArmaProcess(ar,ma = None)
```

```
arma_process.isstationary
```

```
»> yes
```

```
arma_process.arroots
```

```
»> array([1.5 -1.32287566j, 1.5 + 1.3287566j])
```

```
y = arma_process.generate_sample(500) model =
```

```
sm.tsa.ARMA(y,(2,0)).fit(trend = 'nc',disp = 0)
```

```
model.params
```

```
»> array([.69085419, -.2715547])
```


Statsmodels Example: 2

```
import statsmodels.api as sm
import numpy as np

from statsmodels.tsa.arima_process import ArmaProcess
from statsmodels.tsa.arima_model import ARMA

import matplotlib.pyplot as plt

# Simulate the AR series
arparams = np.array([.30, .20, .10]) # AR coefficients
ar = np.r_[1, -arparams] # add zero-lag and negate
ar_process = ArmaProcess(ar)
```

Statsmodels Example: 3

```
ar_process.isstationary
```

```
True
```

```
ar_process.arroots
```

```
array([1.6694 - 2.1640j, 1.6694 - 2.1640j, 1.33874])
```

```
#Generate data and Fit the model
```

```
y = ar_process.generate_sample(250)
```

```
model = ARMA(y, (3, 0)).fit(trend='nc', disp=0)
```

```
model.params # Show the parameters
```

```
(0.310, 0.208, 0.150)
```

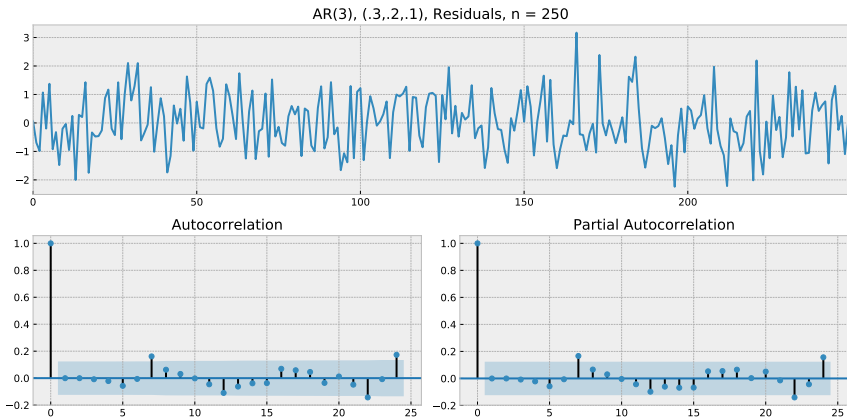
```
plt.plot(model.resid) # Residual Plot
```

```
plt.show()
```

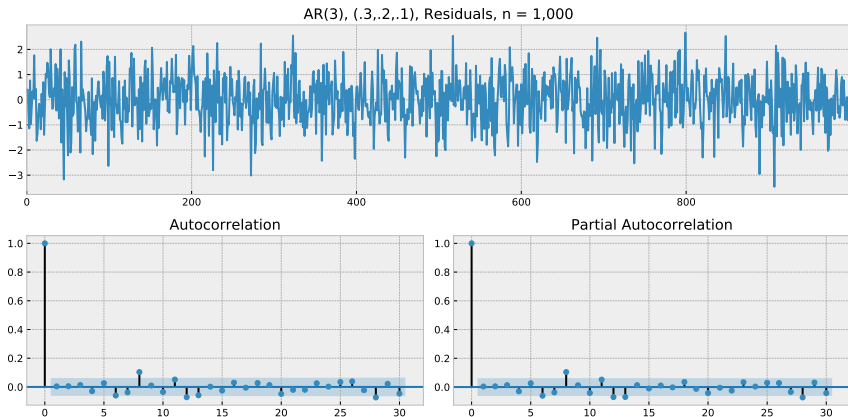
```
plt.acorr(model.resid) #residuals acf plot
```

```
plt.show()
```

Residuals from AR(3), $n = 250$



Residuals from AR(3), $n = 1,000$



Partial Correlation and PACF

Partial Correlation Coefficients: 1

- ▶ In multivariate statistical analysis, one tries to characterize the dependency among a set of variables. For example, consider (X, Y, Z) .
- ▶ The familiar pairwise correlation coefficient, e.g. $\rho(X, Y)$, gives the linearly relationship of X and Y without any consideration of Z .
- ▶ For example, one could start with the joint distribution, $f_{X,Y,Z}(x, y, z)$, integrate out Z leaving $f_{X,Y}(x, y)$ and then compute $\rho(X, Y)$.
- ▶ However, we might want to first eliminate any effects that Z may have on X and Y , and then compute the correlation. This is known as the **partial correlation coefficient**.

Partial Correlation Coefficients: 2

- ▶ The partial correlation of $\rho(X, Y|Z)$ can be found by regressing X on Z and Y on Z separately, then computing the correlation coefficient of the residuals from each of those regressions.
- ▶ Note that Z could be set of random variables, i.e. $\mathbf{Z} = (Z_1, \dots, Z_m)$. One would again regress X and Y on $\mathbf{Z}$ and then find the sample correlation of the two residuals from the regressions.

Partial Autocorrelation Function (PACF): 1

- ▶ The concept of partial correlation can be useful in determining the order of a time series model, especially in the AR process case.
- ▶ Suppose that the data came from a true autoregressive process with order p . We fit an autoregressive model but want to estimate the order.
- ▶ If the true model has order p , then

$$X_t = \mu + \beta_1 X_{t-1} + \cdots + \beta_p X_{t-p} + \epsilon_t.$$

- ▶ The model indicates that the correlation of X_{t-k} and X_t will be 0 conditional on $X_{t-1}, \cdots, X_{t-p}$ for $k > p$ whereas these will be correlated if $k < p$.
- ▶ How can we use this to devise a method to determine the order of an autoregressive process?

Partial Autocorrelation Function (PACF): 2

- ▶ Suppose the true order of the AR process is p for some $p \geq 1$.
- ▶ The k th partial autocorrelation, $\phi_{k,k}$, is estimated by $\hat{\phi}_{k,k}$ in the regression

$$X_t = \phi_{0,k} + \phi_{1,k}X_{t-1} + \cdots + \phi_{k,k}X_{t-k} + \epsilon_{k,t}.$$

- ▶ Note that we contemplate running a series of regressions of the type above as we increase $k = 2, 3, \dots$. The $\epsilon_{k,t}$ have the same structure as the errors in the actual autoregressive process but are different, as are the regression coefficients.
- ▶ If X_t is an AR(p) process, then $\phi_{k,k} = 0$ for $k > p$. Testing that $\phi_{k,k} = 0$ can be used to determine the value of p .

Partial Autocorrelation Function (PACF): 3

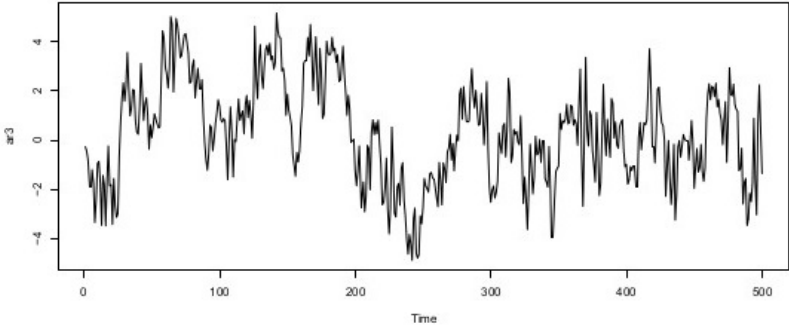
- ▶ Thus, given time series data from an AR(p) process, $\{X_t, 0 \leq t \leq T\}$ One can consider a series of regressions of the form

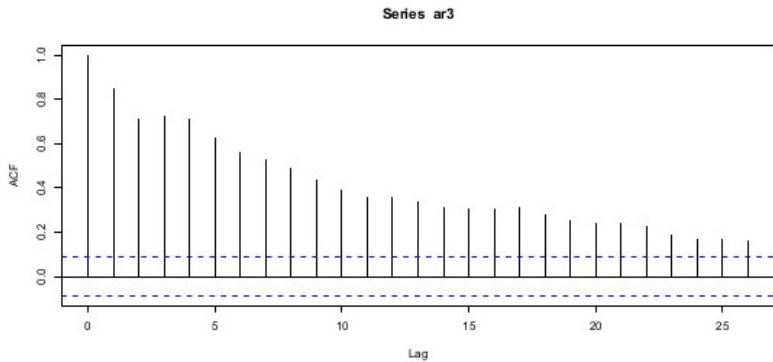
$$X_t = \phi_{0,k} + \phi_{1,k}X_{t-1} + \cdots + \phi_{k,k}X_{t-k} + \epsilon_{k,t},$$

each time testing whether $\phi_{k,k} = 0$.

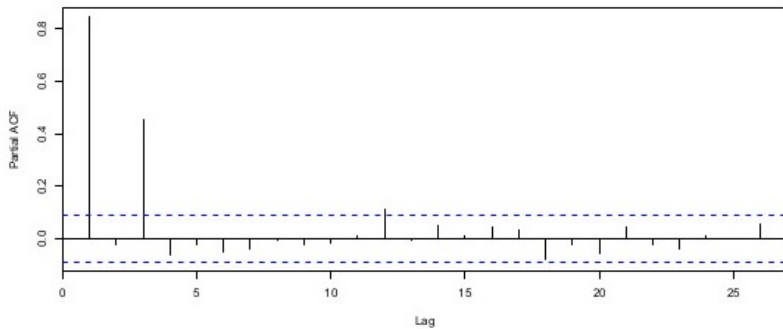
- ▶ The smallest k for which the test allows us to conclude $\phi_{k,k} = 0$ will then give us our estimated order for the process.
- ▶ More generally, one can use the AIC (Akaike Information Criterion) or the BIC (Bayesian Information criterion) based on the log likelihood to determine the order of the autoregression.

AR(3) series

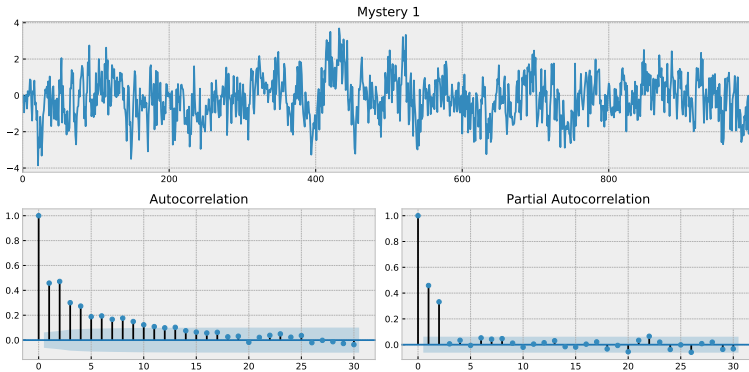




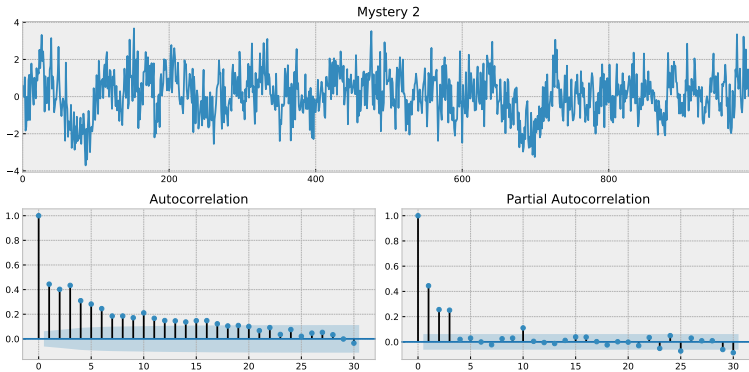
Series ar3



Mystery 1



Mystery 2



Parameter Estimation for AR(p): 1

Starting with an AR(p) process

$$X_t = \sum_{i=1}^p \beta_i X_{t-i} + \epsilon_t,$$

and writing it in vector form

$$\mathbf{X}_t = \boldsymbol{\beta} \mathbf{X}_{t-1} + \epsilon_t,$$

where $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)$ and $\mathbf{X}_{t-1} = (X_{t-1}, \dots, X_{t-p})^T$

- ▶ The equation has a strong resemblance to the familiar regression equation.

Parameter Estimation for AR(p): 2

- ▶ The least squares estimator of β is given by

$$\hat{\beta} = \left(\sum_{t=p+1}^n \mathbf{x}_{t-1} \mathbf{x}_{t-1}^T \right)^{-1} \left(\sum_{t=p+1}^n \mathbf{x}_{t-1} x_t \right).$$

- ▶ In the simple case with $p = 1$,

$$\hat{\beta} = \sum_{t=p+1}^n x_t x_{t-1} / \sum_{t=p+1}^n x_{t-1}^2.$$

- ▶ Most of the standard regression methodology carries over to this autoregressive case, e.g. the asymptotic normality of $\hat{\beta}$ and the asymptotic variance is a function of σ_ϵ and $\gamma(h)$.

Moving Average Processes

Moving Average Processes: 1

- ▶ A moving average process (MA(q)) has the following representation:

$$X_t = \mu + \sum_{i=1}^q \theta_i \epsilon_{t-i} + \epsilon_t,$$

where $\{\epsilon_t, -\infty < t < \infty\}$ is a white noise process.

- ▶ For all $(\theta_1, \dots, \theta_q)$, the $\{X_t, -\infty < t < \infty\}$ process is weakly stationary.
- ▶ Taking expectations across the defining equation, we find $E(X_t) = \mu + \sum_{i=1}^q \theta_i E(\epsilon_{t-i}) + E(\epsilon_t) = \mu + 0 = \mu$.
- ▶ Similarly, computing the variance of each side, $Var(X_t) = \sum_{i=1}^q \theta_i^2 \sigma_\epsilon^2 + \sigma_\epsilon^2 = \sigma_\epsilon^2 (1 + \sum_{i=1}^q \theta_i^2), \forall t$.

Moving Average Processes: 2

- ▶ Thus the MA(q) process has constant mean and variance. Now we must evaluate the autocovariance function. Ignoring the constant μ :

$$\text{Cov}(X_{t+k}, X_t) = \text{Cov}\left(\sum_{i=1}^q \theta_i \epsilon_{t+k-i} + \epsilon_{t+k}, \sum_{j=1}^q \epsilon_{t-j} + \epsilon_t\right)$$

- ▶ X_{t+k} involves $(\epsilon_{t+k}, \dots, \epsilon_{t+k-q})$ while
- ▶ X_t involves $(\epsilon_t, \dots, \epsilon_{t-q})$. If $k > q$
- ▶ If $0 < k \leq q$, then $\gamma(k) = \sigma_\epsilon^2 \sum_{i=0}^{q-k} \theta_i \theta_{i+k}$ which is independent of t , hence an MA(q) is weakly stationary.

Moving Average Processes: 3

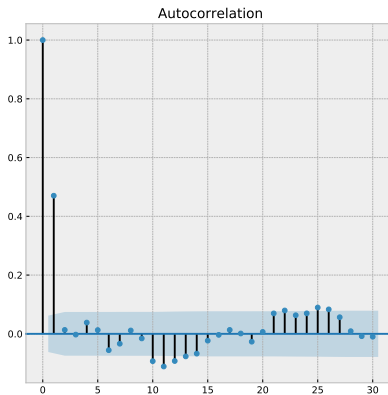
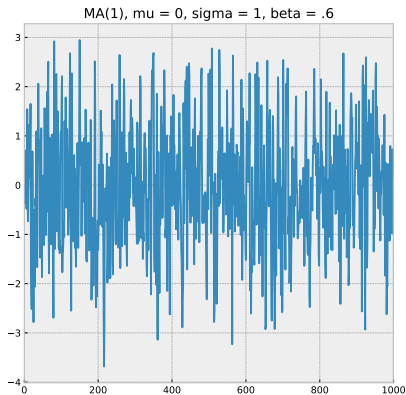
We summarize the properties of an MA(q) process:

- ▶ Weakly stationary for all θ_i , $1 \leq i \leq q$,
- ▶ $E(X_t) = 0$,
- ▶ $\text{Var}(X_t) = \sigma_\epsilon^2(1 + \sum_{i=1}^q \theta_i^2)$, $\forall t$.

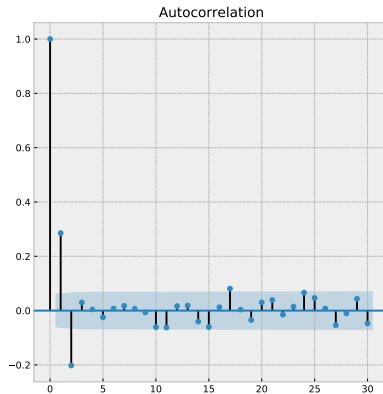
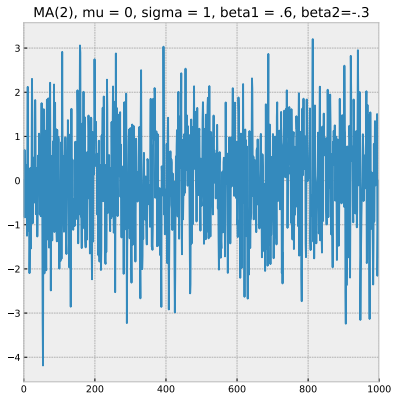


$$\gamma(h) = \begin{cases} \sigma_\epsilon^2 \sum_{i=0}^{q-|h|} \theta_i \theta_{i+|h|} & \text{if } q \leq |h|, \\ 0 & \text{if } q > |h|. \end{cases}$$

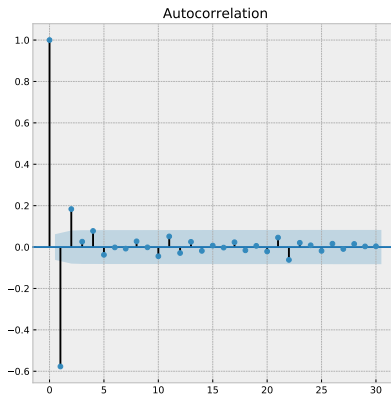
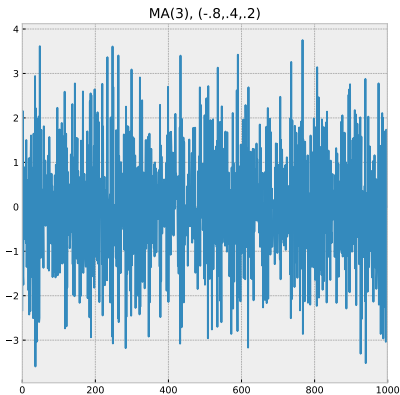
MA(1), (.6)



MA(2), (.6, -.3)



MA(3), (-.8, .4, .2)



MA(q) Analysis

```
# Simulate an MA series
maparams = np.array([.50, -.50]) # MA coefficients
ma = np.r_[1, maparams] # add zero-lag
ma_process = ArmaProcess(ma = ma)

ma_process.isstationary
True

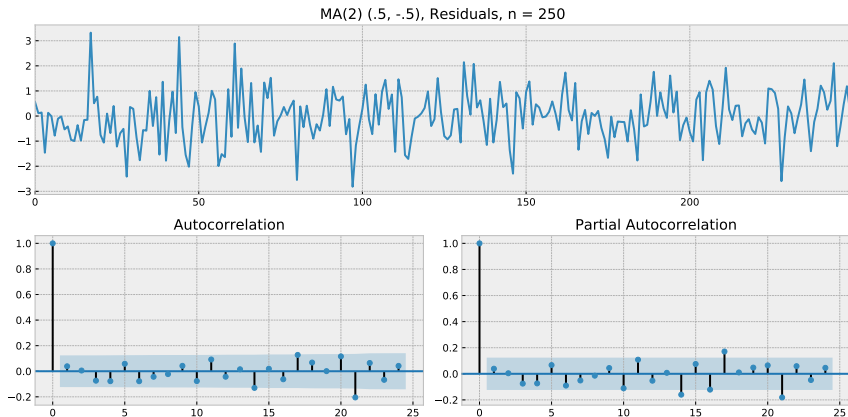
ma_process.isinvertible
True

y = ma_process.generate_sample(250) # Fit the model
model = ARMA(y, (0, 2)).fit(trend='nc', disp=0)

model.params # Show the parameters
array([0.402, -.599]) # n = 250
array([0.504, -.496]) # n = 1,000

plt.plot(model.resid) # Residual Plot
plt.show()
```

Residuals from MA(2), $n = 250$



Residuals from MA(2), n= 1,000

